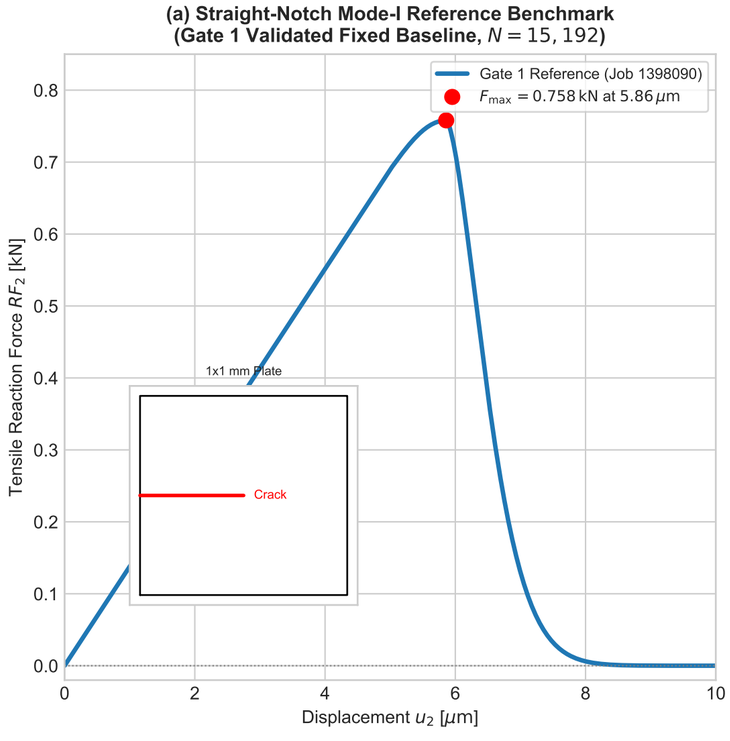
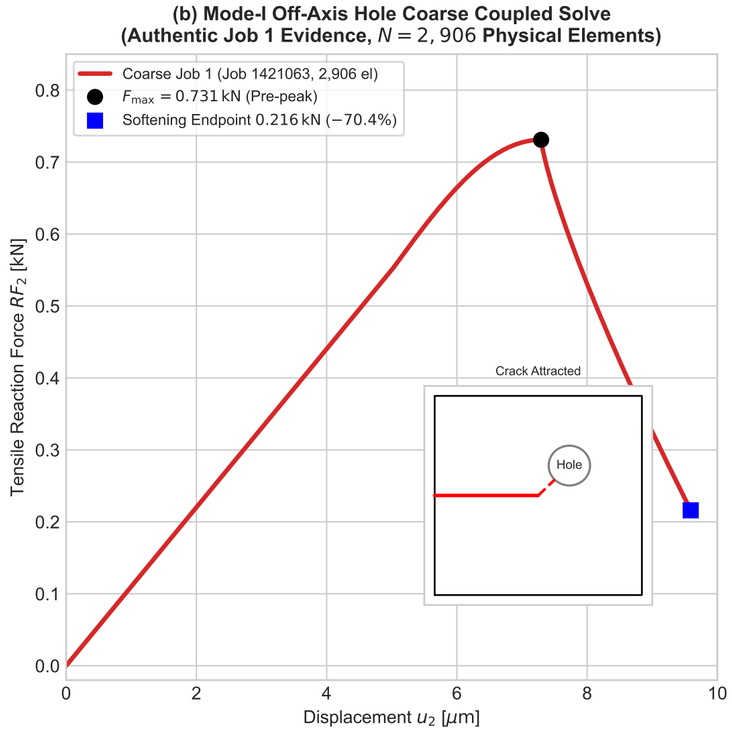
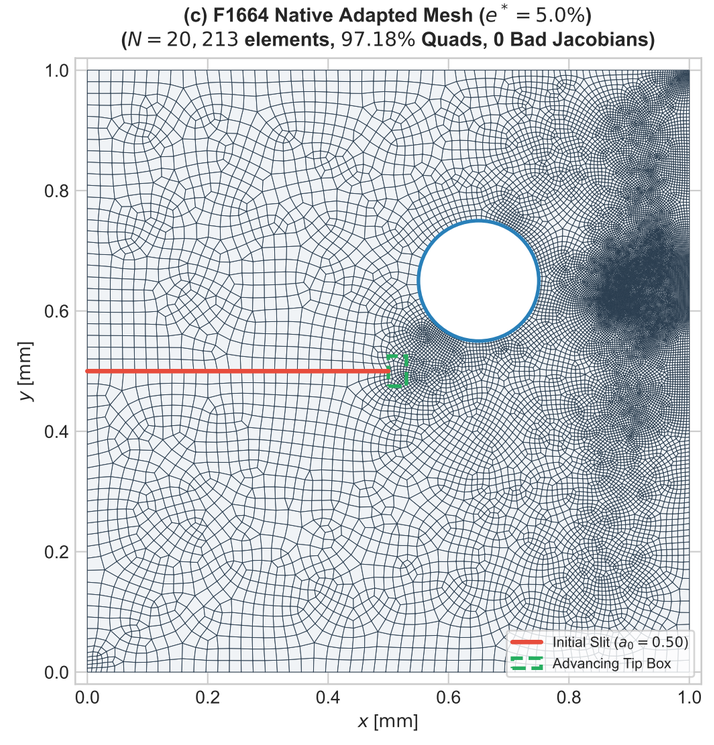
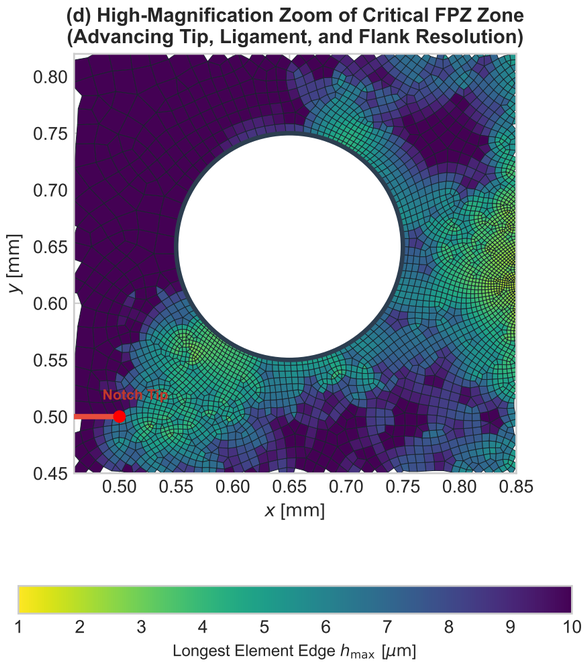
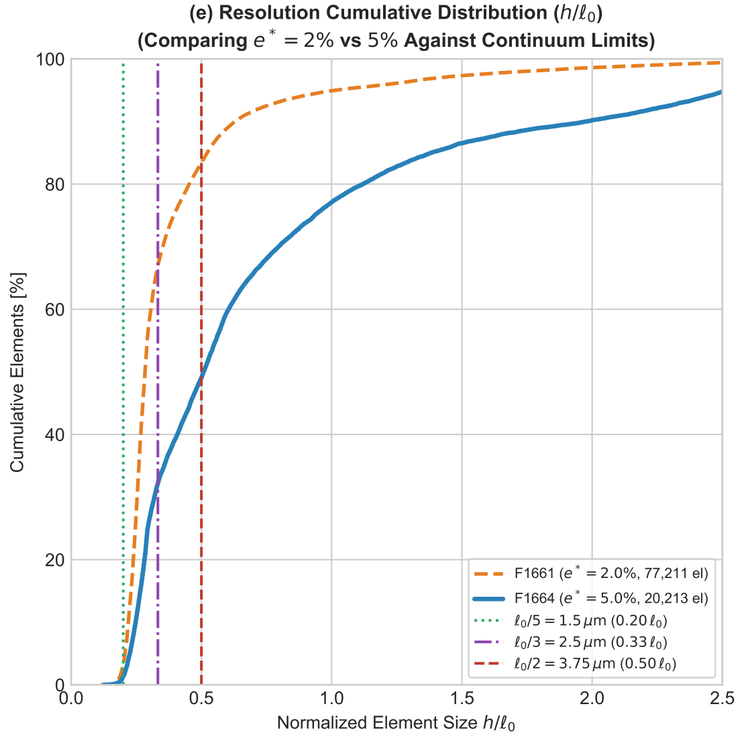
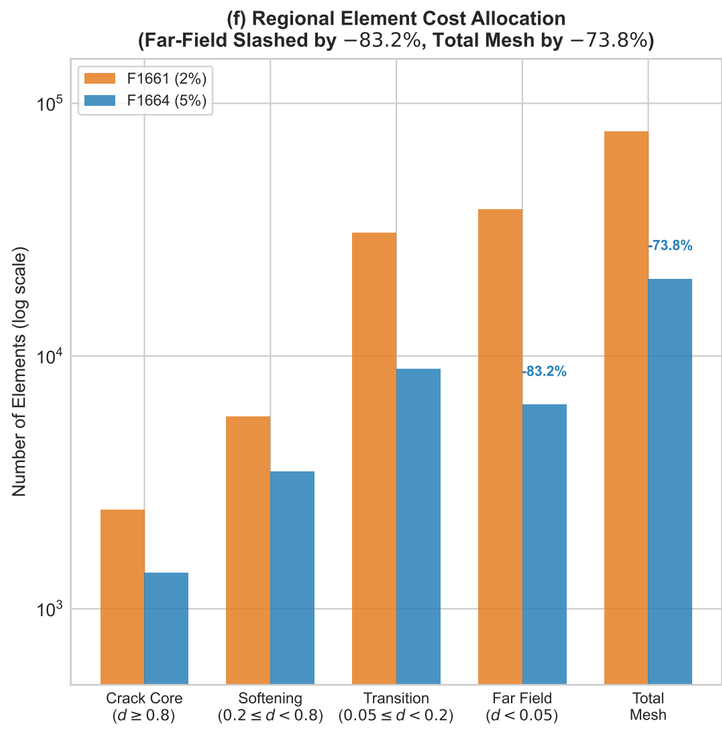

(a) Straight-Notch Mode-I Reference Benchmark
(Gate 1 Validated Fixed Baseline, $N = 15,192$)
Gate 1 Reference (Job 1398090)
$F_{max} = 0.758$ kN at 5.86 μm
1x1 mm Plate
Crack
Tensile Reaction Force RF_2 [kN]
Displacement u_2 [μm]


(b) Mode-I Off-Axis Hole Coarse Coupled Solve
(Authentic Job 1 Evidence, $N = 2,906$ Physical Elements)
Coarse Job 1 (Job 1421063, 2,906 el)
$F_{max} = 0.731$ kN (Pre-peak)
Softening Endpoint 0.216 kN (−70.4%)
Crack Attracted
Hole
Tensile Reaction Force RF_2 [kN]
Displacement u_2 [μm]


(c) F1664 Native Adapted Mesh ($e^* = 5.0\%$)
($N = 20,213$ elements, 97.18% Quads, 0 Bad Jacobians)
Initial Slit ($a_0 = 0.50$)
Advancing Tip Box
y [mm]
x [mm]


(d) High-Magnification Zoom of Critical FPZ Zone
(Advancing Tip, Ligament, and Flank Resolution)
Notch Tip
y [mm]
x [mm]
Longest Element Edge h_{max} [μm]


(e) Resolution Cumulative Distribution (h/ℓ_0)
(Comparing $e^* = 2\%$ vs 5% Against Continuum Limits)
F1661 ($e^* = 2.0\%$, 77,211 el)
F1664 ($e^* = 5.0\%$, 20,213 el)
$\ell_0/5 = 1.5\,\mu$m ($0.20\,\ell_0$)
$\ell_0/3 = 2.5\,\mu$m ($0.33\,\ell_0$)
$\ell_0/2 = 3.75\,\mu$m ($0.50\,\ell_0$)
Cumulative Elements [%]
Normalized Element Size h/ℓ_0


(f) Regional Element Cost Allocation
(Far-Field Slashed by −83.2%, Total Mesh by −73.8%)
F1661 (2%)
F1664 (5%)
-83.2%
-73.8%
Number of Elements (log scale)
Crack Core ($d \geq 0.8$)
Softening ($0.2 \leq d < 0.8$)
Transition ($0.05 \leq d < 0.2$)
Far Field ($d < 0.05$)
Total Mesh